Lifelong Agents: Theory and Practice

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Abstract

Agents are evolving from single-stage task solvers into long-running systems that reason, use tools, maintain memory, and interact with users. As tasks, tools, environments, and safety requirements continually shift, sustaining useful and aligned behavior over an agent's lifetime has become a central challenge. Although recent work has advanced reasoning, alignment, evolution, and evaluation, the literature remains fragmented across capabilities and systems. This tutorial offers a comprehensive introduction to lifelong agents by organizing representative work around five lifecycle stages: foundations, capability development, behavior and alignment, long-term adaptation, and evaluation. We organize recent work on reasoning and planning, tool learning, retrieval and memory, agentic reinforcement learning, safety, multimodal and embodied agents, multi-agent systems, and long-horizon evaluation into a unified lifecycle framework. Through this perspective, attendees will gain a structured overview of the field, a curated reading list, and practical experience in diagnosing failures while developing both insight and foresight for lifelong agent systems. All tutorial materials will be available at our official website.

1 Introduction

Agents have become a central direction in NLP as LLMs are increasingly embedded in systems that reason, use tools, maintain memory, collaborate with other agents, and operate in realistic environments (Wang et al., 2024b; Xi et al., 2025). Rather than treating agents as isolated single-task solvers, this tutorial presents them as long-running systems that must remain useful, reliable, and aligned across sustained interaction.

Research on agents has grown rapidly, but it is often organized around separate capabilities, system designs, and benchmarks (Liu et al., 2024; Zhou et al., 2024; Jimenez et al., 2024). Works on reasoning, memory, alignment, multi-agent and evaluation have produced substantial progress. However, these threads have largely developed in parallel, leaving the field without a unified perspective on how agents should be built, regulated, adapted, and evaluated over their lifetimes.

This tutorial aims to provide a comprehensive, community-facing overview of lifelong agents. We organize representative works from the broader research community around the lifecycle of an agent, covering **foundations**, **capability development**, **behavior and alignment**, **long-term adaptation**, and **evaluation**. Across this structure, we will examine core challenges, representative techniques, common failure modes, and future directions for building agents that remain useful, reliable, and aligned over extended interaction.

This tutorial builds upon the success of our preceding Lifelong Agents workshops, (will be) held at ICLR 2026 and COLM 2026.

2 Tutorial Outline

The tutorial begins with representative failures in current agent systems (Wang et al., 2024b; Xi et al., 2025). A static LLM may rely on outdated parametric knowledge, a tool-augmented agent may make unnecessary tool calls, and a deployed agent may degrade as tools, users, tasks, and safety requirements change (Wang, 2026; Qian et al., 2025). These cases motivate the lifecycle structure of the tutorial. We first introduce what agents are, then study how their capabilities are developed and regulated, how agents adapt over time, and how long-running behavior should be evaluated.

2.1 Foundations of Agents

The first module establishes the conceptual foundation for this lifecycle view. We introduce agents as systems that reason, act, observe, remember, and interact with external environments, and clarify key terms such as tools, environments, memory, trajectories, feedback, and long-horizon interaction. We then connect these concepts to representative settings, including ReAct agents, tool-augmented agents, memory-based agents, multi-agent systems, and embodied agents (Wang et al., 2024b; Xi et al., 2025). The module closes with diagnostic concepts for analyzing common failures, including knowledge boundaries, uncertainty, effort allocation, tool overuse, and tool underuse (Wang, 2026).

2.2 Capability Development in Agents

Building on these foundations, the second module turns to how agents acquire capabilities for sustained interaction. We organize this discussion by the source of capability: internal reasoning and planning for task decomposition and multi-step execution (Wei et al., 2022; Yao et al., 2023); external interaction through tool invocation and composition, retrieval, and API use (Schick et al., 2023; Qin et al., 2024b); and experience-based development through memory, session continuity, and feedback-driven improvement. This organization helps attendees compare how different methods extend agents beyond single-turn generation.

2.3 Behavior and Alignment of Agents

Once agent capabilities have been introduced, the third module shifts from what agents can do to when and how these capabilities should be used. We discuss capability selection, including when to reason, retrieve, call tools, ask clarification questions, or stop; resource control, including reasoning budgets, tool-use budgets, and memory-update decisions (Asai et al., 2024; Qian et al., 2025); and safety constraints, including unsafe tool execution, pre-execution safety checks, and deployment-time alignment concerns (Xia, 2025; Ouyang et al., 2022; Rafailov et al., 2024). This module frames behavioral control as a central issue in deployed agent systems.

2.4 Long-term Adaptation of Agents

The fourth module extends the discussion from within-session behavior to changes that unfold over time. In deployment, tools, users, tasks, and safety requirements may shift, requiring agents to update what they remember, what they can do, and how they interact with their environments. We investigate into works on memory and experience reuse (Park et al., 2023; Packer et al., 2024), skill acquisition and reusable tool compositions (Wang et al., 2024a), and self-improving, embodied, or multi-agent settings (Yang et al., 2024). We will also discuss risks that arise from long-term adaptation, including memory pollution, skill drift, self-reinforcing errors, and unsafe expansion of agent capabilities.

2.5 Evaluation of Agents

The final module examines how the preceding lifecycle should be evaluated. Beyond final-answer accuracy, long-running agents require evaluation of whether they succeed efficiently, remain robust and safe, and stay stable over time. We discuss task success, cost, latency, and resource use (Liu et al., 2024; Mialon et al., 2024); robustness under distribution shift, tool misuse, unsafe execution, and changing environments (Zhou et al., 2024; Xie et al., 2024); and long-horizon properties such as memory quality, adaptation, cross-session stability, and degradation. The tutorial will conclude with a hands-on diagnosis exercise, where participants apply the concepts introduced throughout the tutorial to analyze representative agent trajectories.

3 Type, Audience, and Format

3.1 Tutorial Type and Relevance

This tutorial is proposed as a cutting-edge tutorial in CL/NLP. It addresses a rapidly emerging direction in which language agents are shifting from single-task systems to long-running systems that reason, use tools, maintain memory, adapt to changing conditions, and remain aligned across extended interaction. Recent work on reasoning, tool use, agentic RL, retrieval, memory, safety, and evaluation has grown quickly, but the literature remains distributed across separate capabilities and system designs (Wang et al., 2024b; Xi et al., 2025; Qin et al., 2024a; Zhang et al., 2026). This tutorial provides a structured overview of this growing area by organizing representative work from the broader research community around the lifecycle of language agents: foundations, capability development, behavior and alignment, long-term adaptation, and evaluation. To our knowledge, no prior ACL-family tutorial has provided a lifecycle-based overview of lifelong agents. The closest related tutorial is the EMNLP 2025 tutorial on continual learning of LLMs, which focuses on continual learning and updating of LLMs; our tutorial instead focuses on agent behavior, adaptation, and evaluation over time.

3.2 Target Audience and Prerequisites

The tutorial is intended for researchers, Ph.D. students, advanced master-level students, and industry practitioners working on language agents, tool learning, RAG, dialogue systems, alignment, AI safety, and agentic system deployment. It is self-contained at the conceptual level while focusing on a rapidly developing research area. Expected background includes familiarity with LLMs, prompting, CoT reasoning, RAG, and standard fine-tuning or alignment methods such as SFT, DPO, PPO, or RLHF (Wei et al., 2022; Ouyang et al., 2022; Rafailov et al., 2024). Basic knowledge of RL, Python, or agent frameworks will be useful but not required. We expect **200–300** attendees, based on related community interest: the ICLR 2026 Lifelong Agents Workshop attracted approximately 150 on-site and 400 virtual participants, the SIGIR 2024 Tool Learning Tutorial attracted approximately 150 on-site attendees, and related community talks drew 500–1500 live viewers.

3.3 Format and Schedule

The tutorial will be delivered as three hours of content within a 3.5-hour slot, including a coffee break and QA after each session. As shown in Table 1, the schedule follows the lifecycle structure as introduced in the previous section, and closes with a hands-on diagnosis exercise.

Block	Min	Focus
Welcome	10	Failure cases; tutorial roadmap.
Foundations	30	Agent concepts; settings; diagnostic vocabulary.
Capability Development	35	Reasoning; planning; tools; memory; learning.
Behavior & Alignment	40	Capability selection; budgets; stopping; safety.
Coffee break	30	-
Long-term Adaptation	30	Memory; skills; drift; self-improvement.
Evaluation	35	Multi-dimensional evaluation; failure diagnosis; Q&A.
Total content	**180**	Three-hour tutorial content.
Total slot	**210**	Including coffee break.

Table 1: Proposed schedule for the three-hour tutorial.

4 Specification of the Tutorial

4.1 Relevance and Theme Alignment

This tutorial is designed as a community-facing structured overview rather than a presentation of any single research line. Most of the content will survey representative work from the broader research community, covering reasoning and planning, tool learning, retrieval and memory, behavior and alignment, safety, long-term adaptation, multimodal and embodied agents, multi-agent systems, and long-horizon evaluation (Wang et al., 2024b; Xi et al., 2025; Qin et al., 2024a; Liu et al., 2024). Presenter-led work will be used selectively as case studies and placed alongside related external work to illustrate broader trends, open problems, and methodological trade-offs.

The tutorial aligns with the EMNLP 2026 special theme, “New Missions for NLP Research,” by treating agents as long-running systems rather than static text generators. This perspective connects several core NLP topics, including reasoning, alignment, and evaluation, with deployment challenges that arise when tools, users, tasks, and environments change over time (Zhang et al., 2026). The tutorial therefore highlights a natural new mission for NLP: **studying sustained and adaptive behavior across extended interaction**.

4.2 Participation and Pedagogical Materials

The tutorial supports diversity in representation, expertise, and participation. The presenter team spans multiple countries, institutions, career stages, and areas of expertise. The team brings complementary expertise in agent foundations, tool learning, memory, safety, embodied agents, and multi-agent systems. Participation will be supported through live polling, asynchronous Q&A, a hands-on diagnostic exercise, and the EMNLP hybrid format, so that both in-person and remote attendees can engage with the material.

We will provide classroom-ready slides, an annotated reading list, a hands-on exercise, discussion questions for graduate seminars, and a minimal demo notebook. The reading list will be organized by the tutorial lifecycle, with suggested pre-readings on agent foundations, capability development, behavior and alignment, long-term adaptation, and long-horizon evaluation. Materials will be released at least two months before the tutorial through official website and ACL Anthology.

4.3 Ethics and Venue Compatibility

Safety is treated as a cross-cutting concern for long-running agent systems, especially in the modules on behavior and alignment, long-term adaptation, and evaluation. We will discuss risks related to prompt injection, jailbreak chains, unsafe tool execution, privacy in long-term memory, and multilingual fairness (Xia, 2025; Bai et al., 2022). All examples will be drawn from public benchmarks, and no personally identifying user data will be shown. The tutorial requires only standard conference equipment, including a projector with HDMI or USB-C support, reliable Wi-Fi, microphones, and optional support for live polling. It is compatible with both EMNLP 2026 and AACL-IJCNLP 2026, but with a preference for EMNLP given our workshop's relevance with its special theme.

5 Organizers and Advisory Board

Hongru Wang ([Email] [Website]) is a Research Associate at the University of Edinburgh, advised by Prof. Amos Storkey and Prof. Jeff Z. Pan, and obtained his Ph.D. from CUHK under Prof. Kam-Fai Wong (ACL Fellow). His research builds the *Theory of Agent* and has produced 20+ first/corresponding-author papers at ICML, ACL, EMNLP, NAACL, and ICLR venues, with 2,600+ Google Scholar citations. He co-organized the first Tool Learning Tutorial at SIGIR 2024 (150 onsite attendees) and the first Lifelong Agents Workshop at ICLR 2026 (150 onsite + 400 virtual) as Program Chair—both direct precursors to this tutorial. He serves as Area Chair for NeurIPS 2025/2026 and has delivered 20+ invited talks internationally (UIUC, UCL, HKU, Manchester, EdinburghNLP, AI Time, TechBeat).

Cheng Qian ([Email] [Website]) is a Ph.D. candidate at UIUC's BlenderLab, advised by Prof. Heng Ji. Since 2023, his research has focused on tool learning and agentic AI, particularly lifelong agent learning, alignment, and evolution. His work has appeared in NeurIPS, ACL, and EMNLP and has received over 2500 citations. His representative contributions include ToolRL, accepted to NeurIPS 2025 and adopted by Google, NVIDIA, IBM, and Allen AI; SMART, a foundational study on tool overuse and efficiency; and CREATOR, the first work on tool creation. He has received the Capital One ASKS Center Fellowship, served as an Area Chair for ACL and EMNLP, and is the lead organizer of the ICLR 2026 and COLM 2026 Lifelong Agent Workshops.

Emre Can Acikgoz ([Email] [Website]) is a Ph.D. candidate at UIUC working on conversational agents and tool-overuse mitigation. He co-led SMART, has co-organized the ICLR 2026 Lifelong Agents Workshop, and has prior international teaching experience as a TA for graduate-level NLP courses.

Zhenfei Yin ([Email] [Website]) is a Postdoctoral Researcher at the University of Oxford, advised by Prof. Philip Torr. His research focuses on foundation-model agents, multi-agent systems, self-evolving agents, and embodied agents. His work has appeared at NeurIPS, ICML, ICLR, ACL, and CVPR, with 2,000+ citations and 20,000+ cumulative GitHub stars. His representative contributions include LAMM, OASIS, MARS, and the Agentic RL Survey. He has co-organized international workshops at ICLR, ICML, and CVPR, bringing expertise in agent evolution, multi-agent systems, and embodied intelligence to this tutorial.

Heng Ji ([Email] [Website]) is a Professor at UIUC and an ACL Fellow. Her research focuses on NLP, Data Mining and AI for Science. The awards she received include "AI's 10 to Watch" Award, NSF CAREER award, Google Research Award, IBM Watson Faculty Award, Bosch Research Award, and Amazon AWS Award, ACL 2020 Best Demo Paper Award, and NAACL 2021 Best Demo Paper Award. She has delivered tutorials at ACL, NAACL, EMNLP, KDD, and SIGIR with 3,000+ cumulative tutorial attendees. She is elected as the North American Chapter of the Association for Computational Linguistics (NAACL) secretary 2020-2021.

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